

OmniCore Platform

Technical Specification Document

Version 2.1 | February 2025 | DRAFT - CONFIDENTIAL

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Document status:	DRAFT - pending board approval
Classification:	CONFIDENTIAL - Internal Use Only
Distribution:	Executive team, Tech leads, Investors (on request)

1. System Overview

OmniCore Platform is a multi-tenant, cloud-native Enterprise Resource Planning (ERP) system targeting the manufacturing and distribution sectors in Kazakhstan, Russia, and the broader CIS region. The platform is built on a microservices architecture with 47 independent services deployed on Kubernetes across 3 cloud regions.

Core differentiation: (1) Native AI layer - ML models are first-class citizens, not add-ons; (2) Time-to-value of 3 weeks versus 6-18 months for SAP/Oracle; (3) Low-code BPMN 2.0 process configurator; (4) Full data sovereignty (RU/KZ/EU).

1.1 System Boundaries

The platform boundary includes the following 17 business modules:

- Finance Core (GL, AP, AR, reconciliation, multi-currency)
- Manufacturing Core (MRP II, production orders, BOM, digital twins)
- Warehouse Management System (WMS, RFID, drones, slotting)
- Supply Chain Management (VRP routing, SRM, demand forecasting)
- HR & Payroll (50+ tax jurisdictions, OKR tracking, ATS)
- CRM (360 profile, churn prediction, CPQ, segmentation)
- BI & Analytics (OLAP, NLQ via GPT-4o, real-time dashboards)
- Document Management (DMS, OCR, versioning)
- Project Management (Gantt, milestones, resource planning)
- Asset Management (fixed assets, depreciation, CAPEX tracking)
- Procurement (PO management, vendor portal, e-tendering)
- Quality Control (ISO 9001:2015, inspection plans, NCR tracking)
- Maintenance Management (PPM, work orders, spare parts)
- Transport Management (TMS integration, fleet management)
- Customs & Compliance (HS codes, customs declarations, KZ/RU forms)
- AI Engine (LLM orchestration, embeddings, document intelligence)
- Notification Hub (email, SMS, push, Slack, Teams)

2. Architecture Overview

2.1 Technology Stack Summary

Layer			
API Gateway	Kong	3.6	Rate limiting, JWT validation, routing
Service Mesh	Istio	1.21	mTLS, traffic management, observability
Container Platform	Kubernetes	1.30	EKS (AWS), managed
Backend Primary	Go (Fiber v2)	1.22 / v2.52	All 40 Go microservices
Backend Secondary	Python (FastAPI)	3.12 / 0.111	ML workers, analytics
Backend Legacy	Java (Spring Boot)	21 / 3.3	Legacy ERP adapters
Frontend	React + Next.js	18.3 / 14.2	TypeScript 5.4, SSR+CSR
OLTP Database	PostgreSQL	16.2	Patroni HA, 3 nodes
OLAP Database	ClickHouse	24.3	3 shards x 2 replicas
Cache	Redis	7.2	Sentinel, 3 nodes
Event Streaming	Apache Kafka	3.7	5 brokers, replication=3
Search	Elasticsearch	8.13	Full-text, autocomplete
ML Serving	Triton Inference	24.05	GPU inference, ONNX
IaC	Terraform	1.8	All infra as code
GitOps	ArgoCD	2.11	Declarative deployments
Secrets	HashiCorp Vault	1.16	Dynamic secrets, rotation
Monitoring	Prometheus/Grafana	2.52 / 11.0	Metrics, dashboards, SLOs
Tracing	Jaeger + OTEL	1.57	Distributed tracing
Logging	Loki + Promtail	3.0	Structured logs, 30-day retention

2.2 Service Count and Scaling

The platform comprises 47 microservices (CONFLICT: CTO proposes to consolidate to 32 - decision pending as of Feb 2025, see ADR-002 in architecture.md). Each service is independently deployable with its own CI/CD pipeline, database schema, and scaling policy.

Scaling policy: Kubernetes Horizontal Pod Autoscaler (HPA) is configured for all services with CPU target 70% and memory target 80%. Minimum replicas: 2 (all environments except dev). Maximum replicas: 20 (Go services), 10 (Python/Java services), 6 (ML inference).

2.3 Data Flow Architecture

Primary request path (synchronous, user-facing):

- 1. Client request arrives at CloudFront CDN (TLS termination, static asset cache).
- 2. WAF inspection (AWS WAF + ModSecurity rules, OWASP Top 10 protection).
- 3. Kong API Gateway: JWT validation, rate limit check (1K/10K/unlimited by tier), routing.
- 4. Auth Service: token introspection, RBAC permission check.
- 5. Target microservice processes the request (business logic, DB read/write).
- 6. Response returned with X-Request-ID, X-Trace-ID headers.
- 7. Async: Kafka event published for downstream consumers (BI, audit, notifications).

3. Security Architecture

3.1 Authentication & Authorisation

Identity provider: Keycloak 24 (self-hosted, HA, 3-node cluster). Supports: OAuth 2.0 + OIDC (primary), SAML 2.0 (enterprise SSO), LDAP/Active Directory (on-premise). Token TTL: access token 15 minutes, refresh token 30 days. MFA: TOTP (Google Authenticator / Authy) and WebAuthn/FIDO2 hardware keys.

RBAC model: 5 built-in roles - super_admin, tenant_admin, module_admin, operator, viewer. Custom roles supported via permission matrix. Row-level security enforced via tenant_id predicate on all database queries.

3.2 Data Protection

- Encryption at rest: AES-256-GCM via AWS KMS (automatic key rotation every 365 days).
- Encryption in transit: TLS 1.3 minimum on all connections (clients, service-to-service, DB).
- PII field encryption: SSN, bank account, passport numbers encrypted at field level (AES-GCM, per-tenant key).
- Data masking: PII masked in non-production environments (dev, staging) via automated tooling.
- Secrets management: HashiCorp Vault - database credentials rotated every 24 hours, no static passwords.
- Vulnerability scanning: Trivy (containers) + Snyk (dependencies) in every CI pipeline run.
- SAST: SonarQube - quality gate blocks merge if BLOCKER or CRITICAL issues found.
- Annual penetration testing: external firm, CVSS-based severity classification.

4. Performance Specifications

4.1 Load Targets

Metric		
Total concurrent users	100,000	Across all tenants, all tiers
API RPS (steady state)	50,000 RPS	Normal business hours
API RPS (peak, 3x multiplier)	150,000 RPS	Month-end close, peak season
P99 API response time	<= 200ms	GET /v1/* endpoints under steady load
P99 POST (write) response time	<= 500ms	Write operations with DB commit
Kafka events/second	500,000 ev/s	All topics combined
ClickHouse queries/second	500 concurrent	OLAP analytical queries
Elasticsearch queries/second	5,000 QPS	Full-text search
File export (100K rows Excel)	<= 60 seconds	Background job
Database P99 query latency	<= 20ms	Indexed queries, OLTP only

4.2 Caching Architecture

Multi-layer caching strategy to achieve P99 <= 200ms: (1) CloudFront CDN - static assets (1h TTL), public read endpoints (60s TTL); (2) Kong API Gateway cache (Redis) - read-heavy endpoints, 60s TTL; (3) Application Redis cache - user sessions (24h), RBAC decisions (5min), config (5min), dashboard KPIs (30s); (4) PgBouncer connection pool - eliminates per-request connection overhead.

5. Infrastructure & Cloud

5.1 Multi-Region Deployment

Primary region: AWS us-east-1 (Virginia) - full stack, primary write cluster. Secondary region: AWS eu-central-1 (Frankfurt) - EU GDPR boundary, full stack, EU data residency. Tertiary region: Yandex Cloud ru-central1 (Moscow) - 152-FZ compliance, air-gapped from AWS, RU customer data never leaves this region. Planned Q3 2026: Kazakhtelecom Cloud (Almaty) - KZ government compliance.

5.2 Kubernetes Node Pools (Production)

Pool			
system-pool	m6i.xlarge (4 vCPU, 16 GB)	3	Kong, Istio, ArgoCD, cert-manager
app-pool	m6i.2xlarge (8 vCPU, 32 GB)	24	All 47 business microservices
data-pool	r6i.4xlarge (16 vCPU, 128 GB)	12	Kafka, ClickHouse, PostgreSQL, Redis
ml-pool	g4dn.xlarge (4 vCPU, 16 GB, T4 GPU)	6	Triton inference, ML training
spot-pool	m6i.2xlarge (Spot)	6	Batch jobs, report generation, workers

5.3 Estimated Cloud Costs

CONFLICT: The following cost estimates differ between the CTO and CFO versions of the budget. CTO estimate (for reference only - not approved by CFO): \$31,700/month for cloud (dev+staging+prod). CFO approved budget: \$480,000 total for 18 months = \$26,667/month. CEO requested: \$720,000 for 24 months = \$30,000/month.

Environment		
Production (multi-AZ)	\$18,500	48 nodes, load balancers, managed DBs, CDN
Staging	\$6,200	12 nodes, smaller DB instances
Development	\$4,500	8 nodes, shared DBs
Data lake (S3 + Glacier)	\$1,200	Raw data, backups, ML artifacts
CDN (CloudFront)	\$800	Static assets, API cache
Misc (DNS, IAM, etc.)	\$500	Route53, ACM, CloudWatch
TOTAL	\$31,700 / month	All environments combined

6. ML Platform

6.1 Production Models

The AI Engine (service: ai-engine) orchestrates 6 ML models in production. All models are versioned in MLflow, deployed via Triton Inference Server or FastAPI, and continuously monitored for data drift and accuracy degradation.

Model			
Demand Forecasting	LSTM + Prophet	< 500ms (batch)	Weekly
Churn Prediction	XGBoost	< 50ms (real-time)	Monthly
Anomaly Detection	Isolation Forest	< 20ms (stream)	Weekly
Route Optimization	RL (PPO)	< 2s (per request)	Monthly
Document Classification	BERT-base	< 200ms	Quarterly
Fraud Detection	GNN	< 100ms (real-time)	Weekly

6.2 LLM Integration

The BI & Analytics module uses GPT-4o (OpenAI API) for Natural Language Queries. Users type questions in Russian, Kazakh, or English; the system generates a ClickHouse SQL query, executes it, and returns structured results with a natural language explanation.

Cost estimate per NLQ: \$0.02-\$0.05 depending on query complexity. Monthly cost estimate at 50,000 queries/day: \$30,000-\$75,000. This cost is included in the AI Pack add-on (\$990-\$1,200/month) amortised across tenants.

Local fallback: Llama 3 8B via Ollama, used in air-gapped RU/KZ deployments. Quality trade-off: approximately 20% accuracy reduction on complex analytical queries.

7. Disaster Recovery & Business Continuity

7.1 Recovery Objectives by Tier

Tier			
Enterprise	15 min	5 min	Hot standby (multi-AZ) + async replication
Business	1 hour	30 min	Warm standby + PITR
Starter	4 hours	1 hour	Cold standby + daily backup restore

7.2 Backup Strategy

- PostgreSQL: continuous WAL streaming to S3 (PITR up to 7 days) + weekly full dump.
- ClickHouse: daily incremental + weekly full backup to S3.
- Kafka: MirrorMaker 2 replication to standby cluster in separate AZ.
- Redis: RDB snapshot every 1 hour; AOF for ≤ 1 min RPO on Enterprise.
- Elasticsearch: S3 snapshot every 6 hours.
- S3 object store: cross-region replication enabled (versioning ON).

8. Regulatory Compliance

Russia 152-FZ

Data localisation: all RU personal data stored in Yandex Cloud ru-central1. Air-gapped from AWS. FSTEC certification planned Q2 2026.

Kazakhstan PDP 2023

Personal data stored in Kazakhtelecom Cloud (Almaty). Planned Q3 2026 deployment. Currently deferred.

GDPR

EU customer data in AWS eu-central-1 (Frankfurt). Data retention policies automated. DPA signed with AWS. Right to erasure: automated within 30 days of request.

PCI DSS Level 1

Applicable to SwiftPay integration and billing-service. Annual QSA audit required. Tokenisation via Stripe (no raw card data stored).

ISO 27001

Certification planned Q3 2026. Gap assessment completed Q4 2024. 12 non-conformities identified, remediation in progress.

ISO 9001:2015

Quality management standard enforced in manufacturing module. Inspection records, NCR tracking, corrective actions fully digitised.

9. Budget Summary (Technical Perspective)

WARNING: The following figures are from the CTO's technical budget estimate (Section D of budget_draft.txt). These figures CONFLICT with both the CEO's version (\$6,800,000 total) and the CFO's approved budget (\$4,200,000 total). Final budget is pending board decision.

Cost Category		
Development team salaries	\$164,000	\$3,280,000
Cloud infrastructure (all envs)	\$31,700	\$634,000
Tooling licenses (GitHub, Jira, etc.)	\$7,200	\$144,000
Security & compliance	\$14,200	\$284,000
Marketing & sales	N/A	\$350,000
Reserve (10%)	Varies	\$469,200
TOTAL (CTO estimate)	\$217,100	\$5,161,200

Note: The CTO estimate of \$5,161,200 falls between the CEO's requested \$6,800,000 and the CFO's approved \$4,200,000. All three figures are live and unresolved as of Feb 2025.

Appendix: Document History

Version			
1.0	2024-11-01	Denis Krasnov	Initial draft
1.5	2024-12-10	Denis Krasnov	Added security section
2.0	2025-01-15	Kirill Osipov	Full rewrite with ML section
2.1	2025-02-20	Denis Krasnov	Added cost table, updated service count